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Define Terms

ANLY 540

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```
import pandas as pd

import os
import string
from string import digits
import re

import unicodedata
```

```
#Import the Jobs.csv file
from google.colab import files
uploaded = files.upload()
```

Choose Files Mental-Health-Twitter.csv

```
mhealth = pd.read_csv('Mental-Health-Twitter.csv', encoding = 'utf8')
```

```
##print first 5 rows

mhealth = mhealth.drop('Unnamed: 0', axis =1)
mhealth.head()
```

	post_id	post_created	post_text	user_id	followers	friends	favourites	statuses	retweets	label
0	6.378950e+17	Sun Aug 30 07:48:37 +0000 2015	It's just over 2 years since I was diagnosed w...	1.013187e+09	84	211	251	837	0	1
1	6.378900e+17	Sun Aug 30 07:31:33 +0000 2015	It's Sunday, I need a break, so I'm planning t...	1.013187e+09	84	211	251	837	1	1
2	6.377490e+17	Sat Aug 29 22:11:07	Awake but tired. I need to	1.013187e+09	84	211	251	837	0	1

Next steps: Generate code with mhealth | View recommended plots | New interactive sheet

```
#subset only using first 550 rows to avoid memory usuage issues
mhealth = mhealth.iloc[:550]
```

```
!pip install textacy
import textacy.preprocessing as tprep
```

Show hidden output

```
def normalize(text):
    text = tprep.normalize.hyphenated_words(text)
    text = tprep.normalize.quotation_marks(text)
    text = tprep.normalize.unicode(text)
    text = tprep.remove.accents(text)
    text = tprep.replace.phone_numbers(text)
    text = re.sub(r'http\S+', '', text)
    text = tprep.replace.emails(text)
    text = tprep.replace.user_handles(text)
    text = tprep.replace.emojis(text)
    text = text.lower()
    text = unicodedata.normalize('NFKD', text).encode('ascii', 'ignore').decode('utf-8', 'ignore')
    text = re.sub(r'<[^<>]*>', ' ', text)
    return text
```

```
import nltk
```

```
nltk.download('stopwords')
from nltk.corpus import stopwords
print(stopwords.words('english'))
```

```
['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've", "you'll", "you'd", 'your', 'yours', 'yourself',
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
stopwords = stopwords.words('english')
```

```
#Dataframe
mhealth['clean_post_text'] = mhealth['post_text'].apply(normalize)
mhealth['clean_post_text'] = mhealth['clean_post_text'].map(lambda x: ' '.join([word for word in x.split() if word not in (stopwords)]))

print(mhealth[0:10])
```

```
        post_id                    post_created  \
0  6.378950e+17  Sun Aug 30 07:48:37 +0000 2015
1  6.378900e+17  Sun Aug 30 07:31:33 +0000 2015
2  6.377490e+17  Sat Aug 29 22:11:07 +0000 2015
3  6.376960e+17  Sat Aug 29 18:40:49 +0000 2015
4  6.376960e+17  Sat Aug 29 18:40:26 +0000 2015
5  6.376930e+17  Sat Aug 29 18:26:24 +0000 2015
6  6.376920e+17  Sat Aug 29 18:21:51 +0000 2015
7  6.376890e+17  Sat Aug 29 18:12:59 +0000 2015
8  6.376870e+17  Sat Aug 29 18:04:05 +0000 2015
9  6.376850e+17  Sat Aug 29 17:54:54 +0000 2015

                                           post_text       user_id  followers  \
0  It's just over 2 years since I was diagnosed w...  1.013187e+09         84
1  It's Sunday, I need a break, so I'm planning t...  1.013187e+09         84
2  Awake but tired. I need to sleep but my brain ...  1.013187e+09         84
3  RT @SewHQ: #Retro bears make perfect gifts and...  1.013187e+09         84
4  It’s hard to say whether packing lists are mak...  1.013187e+09         84
5  Making packing lists is my new hobby... #movin...  1.013187e+09         84
6  At what point does keeping stuff for nostalgic...  1.013187e+09         84
7  Currently in the finding-boxes-of-random-shit ...  1.013187e+09         84
8  Can't be bothered to cook, take away on the wa...  1.013187e+09         84
9  RT @itventsnews: ITV releases promo video for ...  1.013187e+09         84

   friends  favourites  statuses  retweets  label  \
0      211         251       837         0      1
1      211         251       837         1      1
2      211         251       837         0      1
3      211         251       837         2      1
4      211         251       837         1      1
5      211         251       837         1      1
6      211         251       837         1      1
7      211         251       837         0      1
8      211         251       837         0      1
9      211         251       837        41      1

                                     clean_post_text
0  2 years since diagnosed #anxiety #depression. ...
1  sunday, need break, i'm planning spend little ...
2              awake tired. need sleep brain ideas...
3  rt _user_: #retro bears make perfect gifts gre...
4  hard say whether packing lists making life eas...
5     making packing lists new hobby... #movinghouse
6  point keeping stuff nostalgic reasons cross li...
7  currently finding-boxes-of-random-shit packing...
8  can't bothered cook, take away way _emoji__emo...
9  rt _user_: itv releases promo video final seri...
```

```
#Lowercase
mhealth['clean_post_text']=mhealth['clean_post_text'].str.lower()

#Remove punctuations and numbers
mhealth['clean_post_text'] = mhealth['clean_post_text'].str.replace(r'[^\w\s]','', regex = True)

mhealth['clean_post_text'] = mhealth['clean_post_text'].str.replace(r'\brt\b', '', regex = True,  case=False) ##remove rt as per correction

mhealth['clean_post_text'] = mhealth['clean_post_text'].str.replace('\d+', '', regex = True)

mhealth['clean_post_text'] = mhealth['clean_post_text'].str.replace(f"[{string.punctuation}0-9]", ' ', regex=True)
mhealth['clean_post_text'].head(5)
```

	clean_post_text
0	years since diagnosed anxiety depression today...
1	sunday need break im planning spend little tim...
2	awake tired need sleep brain ideas
3	user retro bears make perfect gifts great begi...
4	hard say whether packing lists making life eas...

2. Build a word2vec model using the text.
3. Pick 3 target words and find the most similar words to them.
4. Describe what you can infer about the meaning of your three target words.

```
##store in temporary variable and tokenize
!pip install gensim
import gensim

list_ptext = mhealth['clean_post_text'].tolist()

token_ptext = []

for sent in list_ptext:
  temp = gensim.utils.simple_preprocess(sent)
  token_ptext.append(temp)

token_ptext[10]
```

Show hidden output

```
token_ptext[15]
```

```
['much', 'stuff', 'way', 'way', 'much', 'massive', 'purge', 'way']
```

```
token_ptext[20]
```

```
['get', 'lie', 'little', 'longer', 'cat', 'buddy', 'emoji']
```

```
from gensim.models import Word2Vec
model = Word2Vec(token_ptext,
        vector_size=100,
        window=6,
        min_count=2,
        workers=4)
```

```
model.wv.most_similar("anxiety")
```

```
[('new', 0.36408913135528564),
 ('behind', 0.31389597058296204),
 ('men', 0.3136141300201416),
 ('heart', 0.29597121477127075),
 ('suffering', 0.2915388345718384),
 ('school', 0.28957700729370117),
 ('things', 0.26712241768836975),
 ('head', 0.25641462206840515),
 ('htt', 0.25463661551475525),
 ('high', 0.25386324524879456)]
```

```
model.wv.most_similar("depression")
```

Show hidden output

```
model.wv.most_similar("love")
```

Show hidden output

```
model2 = Word2Vec(token_ptext,
        vector_size=100,
        window=6,
        min_count=2,
        workers=4,
        sg = 1)
```

```
target_similar = []
target_similar.append(model.wv.most_similar("anxiety", topn=10))
target_similar.append(model2.wv.most_similar("anxiety", topn=10))

sim_mhealthtext = pd.DataFrame(target_similar).T
sim_mhealthtext
```

	0	1
0	(new, 0.36408913135528564)	(new, 0.47176212072372437)
1	(behind, 0.31389597058296204)	(men, 0.39243754744529724)
2	(men, 0.3136141300201416)	(things, 0.36577877402305603)
3	(heart, 0.29597121477127075)	(behind, 0.36045610904693604)
4	(suffering, 0.2915388345718384)	(mental, 0.34287944436073303)
5	(school, 0.28957700729370117)	(little, 0.34167563915252686)
6	(things, 0.26712241768836975)	(school, 0.34026482701301575)
7	(head, 0.25641462206840515)	(heart, 0.3392809331417084)
8	(htt, 0.25463661551475525)	(high, 0.32108035683631897)

Next steps: Generate code with `sim_mhealthtext` View recommended plots New interactive sheet

```
target_similar = []
target_similar.append(model.wv.most_similar("depression", topn=10))
target_similar.append(model2.wv.most_similar("depression", topn=10))

sim_text = pd.DataFrame(target_similar).T
sim_text
```

	0	1
0	(like, 0.32929253578186035)	(like, 0.39807432889938354)
1	(even, 0.25466105341911316)	(new, 0.34059035778045654)
2	(coming, 0.24895967543125153)	(come, 0.31138938665390015)
3	(marriage, 0.2385709285736084)	(even, 0.30591800808906555)
4	(die, 0.23031479120254517)	(looking, 0.3021000921726227)
5	(effect, 0.22902972996234894)	(coming, 0.29939940571784973)
6	(school, 0.2286747843027115)	(know, 0.29391980171203613)
7	(come, 0.2278418242931366)	(busy, 0.2847250998020172)
8	(new, 0.22481627762317657)	(school, 0.28191855549812317)

Next steps: Generate code with `sim_text` View recommended plots New interactive sheet

```
target_similar = []
target_similar.append(model.wv.most_similar("love", topn=10))
target_similar.append(model2.wv.most_similar("love", topn=10))

sim_posttext = pd.DataFrame(target_similar).T
sim_posttext
```

	0	1
0	(felt, 0.3102179765701294)	(user, 0.3541382849216461)
1	(video, 0.2767421305179596)	(work, 0.3192923665046692)
2	(wondered, 0.2536873519420624)	(felt, 0.31188419461250305)
3	(week, 0.24264563620090485)	(video, 0.3096488118171692)
4	(say, 0.232512429356575)	(wondered, 0.2985056936740875)
5	(crocheting, 0.2302929311990738)	(week, 0.29762133955955505)
6	(true, 0.2233177274465561)	(busy, 0.29082608222961426)
7	(invisible, 0.22023680806159973)	(true, 0.2866789400577545)
8	(real, 0.2187744677066803)	(say, 0.279493123292923)

Nex Generate code with `sim_posttext` View recommended plots New interactive sheet

Describe what you can infer about the meaning of your three target words.

Due to past assignments and works I have done using this data, I picked three target words I have come across while working on this dataset:

- Anxiety
  - Themes I got from "Anxiety" seem to revolve around emotions, personal relationships, feeling, and mental health challenges.
  - Words like "men," "behind," "heart," "mental," "school," and "high" seem to suggest things that may contribute to anxiety, hence their similarities and association with the word.
- Depression
  - I got from "Depression" centered around negative and stressful emotional states and surroundings.
  - Words like "die," "survivors," "busy," "effect," "marriage," and "school" could be associated with negative emotions/interactions and poor mental health (such as depression). However, this also depends on the context.
- Love
  - The main themes I got from "Love" is based on feelings and actions. The associated or similar words for love do suggest that it is closely tied to how a person is feeling and what they are doing.
  - Words like "felt" "true" "real" "thought" "help" and even "crocheting." There are other similar words such as "invisible" and "busy" which may also allude to Love, if seen in that context.

Overall, out of the three target words, Love is the target word that has the most near positive and neutral themes. Which is as expected, since the word Love, often, revolves around positive emotions or actions. Depression and Anxiety are both associated with similar wards that alludes to negative or near negative emotional state or action.

For our Word2vec model outputs, we can see the similarity scores between the target words and the words associated with them. A high similarity score would indicate that the word is strongly associated with the target word. For example, in the context of this research, "felt" "wondered" "user" and "work" appear under the Love target word and have higher similarity scores which indicates that it is strongly associate to the target word.